

PDF Audio Reader

Project Overview:

PDF Audio Reader is a desktop application developed using Python that converts text-based PDF documents into speech. The application provides a simple graphical user interface (GUI) that allows users to select a PDF file from their computer and listen to its contents. This project was designed to improve accessibility and provide an alternative way of consuming textual information. The application is especially useful for users who prefer listening to documents rather than reading them, such as students, professionals, and individuals who multitask while studying or working.

Problem Statement:

Reading lengthy PDF documents can be time-consuming and tiring. Although many documents are available in PDF format, users often require an easier way to access the information without continuously looking at a screen. The objective of this project is to provide a simple solution that transforms PDF text into audio.

Objectives:

- To develop a user-friendly desktop application for reading PDF documents aloud.
- To provide a simple graphical interface for selecting and managing PDF files.
- To enable users to start and stop audio playback as required.
- To demonstrate the integration of Python libraries for file processing and text-to-speech conversion.

Features:

- Select PDF files using a graphical file browser.
- Convert text-based PDF content into speech.
- Start and stop reading functionality.
- Lightweight and easy-to-use interface.
- Works offline without requiring an internet connection.

Technologies Used:

- Programming Language - python

Libraries and Modules:

- Tkinter – Used to create the graphical user interface.
- PyPDF2 – Used to extract text from PDF documents.
- pyttsx3 – Used for offline text-to-speech conversion.
- threading – Used to maintain GUI responsiveness while reading PDF content.

Deployment Tool:

- PyInstaller – Used to convert the Python application into a standalone Windows executable file (.exe).

System Workflow:

1. The user launches the application.
2. The user clicks the **Open PDF** button.
3. The application opens a file dialog to select a PDF document.
4. The selected PDF is processed using the PyPDF2 library.
5. The extracted text is sent to the pyttsx3 text-to-speech engine.
6. The application reads the content aloud.
7. The user can stop the reading process at any time using the **Stop Reading** button.

Limitations:

- The application supports only text-based PDFs.
- Scanned or image-based PDFs are not supported in the current version.
- The quality of speech depends on the voices installed on the operating system.

Future Enhancements:

- Integration of OCR (Optical Character Recognition) to support scanned PDFs.
- Multiple voice options.
- Adjustable speech speed controls.
- Reading progress indicators.
- Support for exporting speech as audio files.

Conclusion:

The PDF Audio Reader project demonstrates the practical application of Python in developing assistive technologies. By integrating PDF processing and text-to-speech functionalities within a graphical interface, the application provides users with an efficient and accessible way to consume textual content. This project also highlights the importance of creating software solutions that improve usability and accessibility for a wider audience.